

PAPER_928: GW190425 Wavelength Phonon Correction

Author: Daniel T. Murphy - Star Magic / UQFF Framework **Date:** 2026-04-11 **Session:** 211 **Source:** S_{Cm} phonon gap implementation (ns_phonon_gw190425_wstp.py) **Calculator:** GW190425WavelengthPhononCorrectionCalc (CP4 #512) **CVW:** v2.0.0 compliant

Abstract

Derives the phonon-corrected gravitational wave wavelength: $\lambda_{\text{UQFF}} = \lambda_{\text{GR}} * (1 - F_{\{U,Bi\}}/F_U * \Phi_{\text{norm}})$, where $\Phi_{\text{norm}} = \Phi/\Phi_0$ is the normalized phonon modulation. This implies a refractive-index-like effect in the UQFF vacuum structure, where phonon-vacuum coupling modifies the effective propagation speed of gravitational waves. At resonance ($\omega = \omega_{\text{SCm}}$), $\Phi_{\text{norm}} = S_{26} \sim 0.57$ and the wavelength correction is proportional to the buoyancy ratio $F_{\{U,Bi\}}/F_U$, providing a frequency-dependent GW dispersion relation unique to the UQFF framework.

1. Core Equations

Section A: Lagrangian

$\lambda_{\text{UQFF}} = \lambda_{\text{GR}} * (1 - F_{\{U,Bi\}}/F_U * \Phi_{\text{norm}})$
 $\lambda_{\text{GR}} = c / f_{\text{GW}}$
 $\Phi_{\text{norm}} = \Phi_{\{1.25\text{THz}\}}(\omega) / \Phi_0$
 $\Delta\lambda = \lambda_{\text{GR}} - \lambda_{\text{UQFF}} = \lambda_{\text{GR}} * F_{\{U,Bi\}}/F_U * \Phi_{\text{norm}}$

Section B: VDS/DVP/BH Number Systems

Effective refractive index: $n_{\text{UQFF}} = 1 / (1 - F_{\{U,Bi\}}/F_U * \Phi_{\text{norm}})$
GW dispersion: $\omega^2 = k^2 * c^2 * (1 - F_{\{U,Bi\}}/F_U * \Phi_{\text{norm}})^2$
Phase velocity: $v_{\text{ph}} = c * (1 - F_{\{U,Bi\}}/F_U * \Phi_{\text{norm}})$

Section SM: SM Anchors

$c = 2.998e8 \text{ m/s}$
 $f_{\text{GW}} = 20\text{-}300 \text{ Hz}$ (LIGO band)
 $F_{\{U,Bi\}}/F_U = 0.6$ (typical buoyancy ratio)
 $\Phi_0 = 10^{20}$ (peak phonon amplitude)

2. Parameters

Parameter	Default	Description
f_{GW}	300 Hz	GW frequency
F_{UBi}	0.6 N	Buoyancy force
F_U	1.0 N	Unified force
ω	ω_{SCm}	Phonon frequency

3. Key Results

f_GW (Hz)	lambda_GR (m)	Delta_lambda (m)
20	1.499e7	$\sim F_{\text{ratio}} * \Phi_{\text{norm}} * \lambda$
100	2.998e6	proportional
300	9.993e5	proportional
1000	2.998e5	proportional

4. Physical Interpretation

The wavelength correction implies that gravitational waves propagate through the UQFF vacuum with an effective refractive index $n > 1$, analogous to electromagnetic waves in a dielectric medium. The phonon-vacuum structure acts as a gravitational medium, with the buoyancy ratio $F_{\{U,Bi\}}/F_U$ controlling the coupling strength. This predicts a frequency-independent fractional wavelength shift (unlike electromagnetic dispersion), which could be tested by comparing arrival times of multi-frequency GW components from the same source.

5. References

- PAPER_927: GW190425 phonon-suppressed strain
- PAPER_916: GW190425 mass-gap phonon suppression
- ns_phonon_gw190425_wstp.py: 5-class standalone module
- WSTP expression #28: lambda_UQFF GW190425